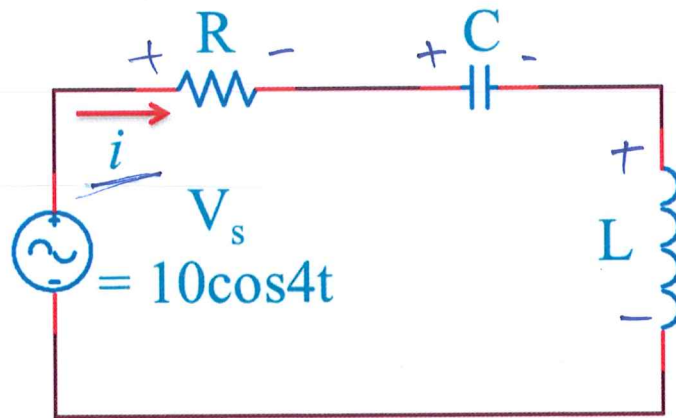


L#26: Phasor and circuit elements

How to analyze an AC circuit?



KVL

$$-10 \cos 4t + iR + \frac{1}{C} \int_0^t i \, dt + L \frac{di}{dt} = 0. \quad (1)$$

$$40 \sin 4t + R \frac{di}{dt} + \frac{i}{C} + L \frac{d^2 i}{dt^2} = 0$$

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} = -40 \sin 4t$$

2nd order differential
equation

Phasor

A phasor is a Complex number that represents the Amplitude and phase of a sinusoid (Cosine).

AC signal & phasor

Euler's identity: $e^{\pm j\phi} = \cos\phi \pm j \sin\phi$

$$\operatorname{Re}\{e^{j\phi}\} = \cos\phi$$

$$\operatorname{Im}\{e^{j\phi}\} = \sin\phi$$

$$e^{j\omega t} = \cos\omega t + j \sin\omega t$$

$$\cos\omega t = \operatorname{Re}(e^{j\omega t})$$

$$v(t) = V_m \cos(\omega t + \phi)$$

Understand the cosine function:

$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

$$e^{j(\omega t + \phi)} = \cos(\omega t + \phi) + j \sin(\omega t + \phi)$$

$$V_m e^{j(\omega t + \phi)} = \underline{V_m \cos(\omega t + \phi) + j V_m \sin(\omega t + \phi)}$$

$$\Rightarrow V(t) = V_m \cos(\omega t + \phi) = \operatorname{Re}(V_m e^{j(\omega t + \phi)})$$

$$= \operatorname{Re}(\underline{V_m \cdot e^{j\phi} \cdot e^{j\omega t}})$$

$$= \operatorname{Re}(\bar{V} \cdot e^{j\omega t})$$

$$\begin{aligned} \bar{V} &= V_m e^{j\phi} = \underline{V_m \angle \phi} \\ &= x + jy \end{aligned}$$

Phasor transformation

$$\underline{v(t) = V_m \cos(\omega t + \phi)} \longleftrightarrow V = V_m \angle \phi$$

- ① V_m : positive,
- ② Must be cosine
- ③ $-180^\circ \leq \phi \leq 180^\circ$

Time domain representation

Phasor domain representation

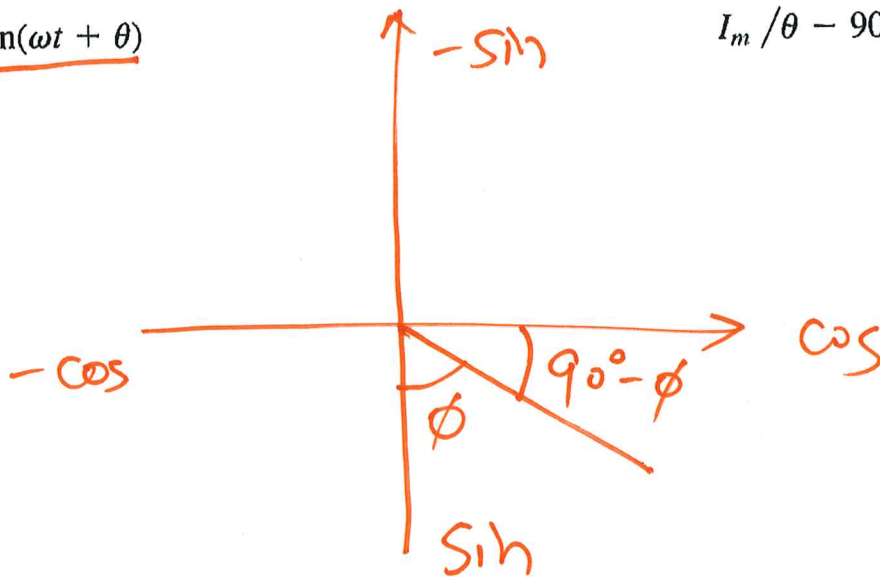
$$\underline{V_m \cos(\omega t + \phi)}$$

$$\underline{V_m \sin(\omega t + \phi)}$$

$$\underline{I_m \cos(\omega t + \theta)}$$

$$\underline{I_m \sin(\omega t + \theta)}$$

$$\begin{aligned} & \rightarrow V_m \cos(\omega t + \phi - (90^\circ - \phi)) \quad \underline{V_m \angle \phi} \\ & = V_m \cos(\omega t + \phi - 90^\circ) \quad \underline{V_m \angle \phi - 90^\circ} \\ & \quad \quad \quad \underline{I_m \angle \theta} \\ & \quad \quad \quad \underline{I_m \angle \theta - 90^\circ} \end{aligned}$$

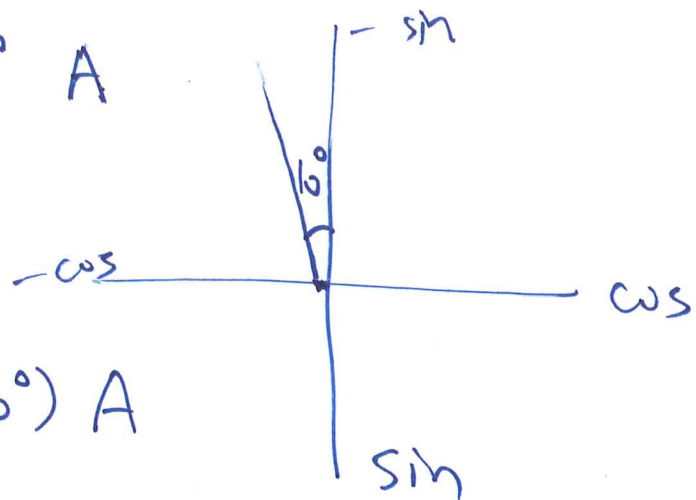


Example #1: Express these sinusoids as phasors, and then get the sum of the two current.

$$i_1 = 7 \cos(10t + 40^\circ) \text{ A}$$

$$i_2 = -4 \sin(10t + 10^\circ) \text{ A}$$

$$\bar{I}_1 = 7 \angle 40^\circ \text{ A}$$



$$i_2 = 4 \cos(10t + 100^\circ) \text{ A}$$

$$\bar{I}_2 = 4 \angle 100^\circ \text{ A}$$

$$\bar{I}_1 + \bar{I}_2 = 7 \angle 40^\circ + 4 \angle 100^\circ$$

$$= \sqrt{93} \angle 61^\circ \text{ A}$$

$$= 4.67 + j8.44 \text{ (A)}$$